

PIPE JOINT INSPECTION APP

Brief Overview

We are working on a pre-compliance tool for S104 drainage adoption. It will be a simple offline app (currently maybe a website at trial level) used on construction sites to diagnose photos of pipe joint (foul and surface water) during installation.

PURPOSE

The purpose is to increase chances of adoption and help contractors(sub) detect installation issues *before backfilling*, reducing future defects.

Core workflow:

User selects project → selects manhole → takes guided joint photos → app automatically measures joint gap → app assigns tolerance status → user exports evidence pack.

3. MVP Scope (Strict)

The app will support **joint inspection using photos only**.

No video.

No cloud accounts.

No live streaming.

No AI defect detection beyond gap measurement.

Core Feature List

-Project Creation

User can create a project.

Fields: (text field)

- Project Name (required)
- Site Name (optional)
- Date (auto-generated)

-Manhole Creation

Inside a project, user creates Manholes.

Field:

- Type (foul/ surface. This should a drop down)
- Manhole ID (required) (this should be a text field)
- Meter run (support decimal)

Pipe types, diameter and length

- 150 mm (clay) – 1.75m
- 225 mm (clay) – 2m
- 300 mm (concrete)- 2.6m
- 450 mm (concrete)- 2.6m
- 600 mm (concrete) – 2.6m
- 900 mm (concrete)

pipe Joint Inspection Workflow

User selects Manhole → taps **Start Joint Inspection**

The app enters upload mode from gallery. (support pictures only)

It must support multiple picture upload and recognise it in the order of upload. It must also be able to annotate each joint as 1-2, 2-3 and so forth.

I have used AI to generate this for better explanation

- Automatic Joint Gap Measurement

This is the core function of the app/website that we must get right.

7.1 Measurement Method

The app must automatically estimate the gap between two pipes at the joint using computer vision.

Method:

1. Detect circular pipe opening in image.
2. Use known pipe diameter to calculate image scale.
3. Detect edges of both pipe ends.
4. Measure gap in pixels.
5. Convert to millimetres.
6. Display measured gap to user.

Must work offline.

Accuracy expectation:

±2–3 mm.

7.2 Manual Override

User must be able to:

- Adjust gap value manually if needed.
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8. Gap Tolerance Classification

The app must classify the gap using the following rules:

Gap (mm) Status Label

< 3 mm Review Gap too small

3 – 15 mm Pass Within tolerance

15 – 25 mm Review Larger than typical

Gap (mm)	Status	Label
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> 25 mm	Fail	Excessive gap
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8.1 On-screen Result Example

After measurement:

Display:

Measured gap: 6 mm

Status: PASS

or

Measured gap: 22 mm

Status: REVIEW

or

Measured gap: 32 mm

Status: FAIL

Include note:

“Guidance only – not a formal adoption assessment.”

Final report in this format

Required fields:

- Project name
- Manhole ID
- Date/time
- Pipe type
- Pipe diameter
- Number of joints captured
- Gap value (mm)
- Gap status
- User notes (optional)

Store as JSON locally.

11. Inspection Summary Screen

After inspection: overall report if good, bad or review. Specify the bad/review with appropriate number.